When accounting information arrives too early: a reproducible audit of publication timing and survivor selection in Borsa Istanbul equity prediction

Working manuscript - bounded existing-evidence draft. Research support only; not investment advice. Selection-effect measurement is pending. Evidence notes link to the accompanying citation crosswalk; they are retained for review before journal formatting.

Abstract

Financial prediction studies can respect chronological train-test splits while still using information unavailable at the prediction date. We audit a completed Borsa Istanbul equity-ranking study for publication-time leakage and retrospective survivor selection. The frozen capstone universe comprises 81 firms, with 40 reported; it was selected in 2026 from then-current listings under an undocumented rule. [N01] The original equal-weight baseline had a pooled Spearman information coefficient (IC) of +0.150 (permutation p = 0.017). The availability-aware re-evaluation reports +0.031 (p = 0.63; 95 % bootstrap interval −0.100 … +0.162). [N02] The correction combines publication-time alignment with changes to the return window, return source, and admissible rows; this contrast does not isolate a causal effect of timing. No machine-learning model is distinguishable from the within-year null after the reported family adjustment. [N03] The reported minimum detectable |IC| is 0.185 at 80 % power, limiting the interpretation of the null. [N04] Survivor conditioning is established by the selection frame, but neither its performance effect nor additional realised-return-based selection has been established. A frozen-design comparison with a survivor-inclusive point-in-time universe is planned and has not run. The contribution is a reproducible audit connecting information availability, sample construction, executable guards, and restrained inference. The corrected results are post hoc and descriptive, not clean confirmatory evidence of predictive ability.

Keywords: look-ahead bias; publication lag; survivorship; financial machine learning; point-in-time evaluation; reproducibility.

Introduction

A chronological split is necessary for financial prediction, but it does not establish that a historical feature vector could have been observed when a prediction was supposedly made. A financial statement describes an accounting period; its publication occurs later. Likewise, a contemporary list of firms does not describe the population that was eligible at an earlier formation date. Retrospective data assembly can therefore encode future information in both the values supplied to a model and the identities admitted to its sample.

This paper examines these problems in the frozen FinanceIQ capstone, a study of annual cross-sectional stock-return ranking. Its original headline was an apparent weak signal from a simple baseline. A subsequent audit found that fiscal-year statements had been used before publication, alongside incompatible market-value adjustment bases and pre-listing labels. The revised evaluation retained the historical record and replaced its predictive interpretation with a bounded null finding. The question here is how to make that revision auditable without overstating what a bundled correction identifies. [N05]

We distinguish publication-time look-ahead from selection-time look-ahead. The former asks whether each input was public before prediction and before the target window. The latter asks whether sample eligibility was determined using information from a later date. Choosing then-listed survivors for a historical panel introduces the latter defect even without proof that firms were explicitly ranked on realised returns. These mechanisms require different evidence and different remedies. Correct feature timing cannot restore firms omitted by a survivor-conditioned selection frame.

The contribution is an auditable case study, rather than a new principle of financial data lagging. It brings together a preserved original evaluation, a registered-before-own-outcomes but post-hoc correction, an availability registry and machine-checked guard, and an explicit account of unresolved selection. The completed numerical evidence concerns the timing-and-data correction package. Selection-effect measurement remains a clearly marked incomplete component; we do not claim that both effects have been quantified. This distinction matters for replication: a documented defect can warrant withdrawing an interpretation even when its individual contribution to the observed performance change is unidentified.

Related work and conceptual framework

Accounting-information lags are established practice. Fama and French (1992), Section I.A, match accounting observations to subsequent return tests with a conservative minimum six-month gap. [L01] That convention motivates attention to availability, but it is not a universal legal filing deadline. A local evaluation must respect its market's actual publication rules and evidence. Drobetz and Otto (2021) provide a related machine-learning asset-pricing setting in which lagging fundamentals and sequential training, validation, and testing are explicit design choices. [L02] We do not import their economic results into this sample.

Kapoor and Narayanan (2023) treat leakage as a broader scientific-reproducibility problem and provide a taxonomy for checking it. [L03] Our case separates defects in feature availability, sample eligibility, and measurement. A feature may be correctly associated with an accounting year yet unavailable at the decision date. A sample can contain accurately measured values yet condition inclusion on later survival. An adjustment-basis error can alter inputs without being a publication-lag error. A useful audit must test these distinctions instead of treating all data problems as one interchangeable form of leakage.

The target-trial framework offers a methodological analogy. Hernán and Robins (2016), in their discussion of defining time zero, emphasize that eligibility must hold at baseline and outcomes must begin afterward. [L04] Misaligning eligibility, assignment, and follow-up is the clinical setting in which immortal-time problems arise. The financial counterpart considered here is an alignment question: which firms were eligible, which information was public, and when did the evaluated return begin? This analogy does not make a stock-ranking exercise a randomized trial or identify a causal effect of an audit intervention.

Brown et al. (1992) show why survivor truncation can create apparent performance persistence in a performance-study setting. Shumway (1997) documents a distinct problem involving missing delisting returns. [L05] These papers motivate separate checks for membership inclusion and exit-outcome measurement. They do not supply a transferable correction factor for this capstone's rank IC. In particular, knowing that the historical sample is survivor-conditioned does not reveal the direction of its effect on cross-sectional ranking performance.

Multiple testing adds another layer. White (2000), Hansen (2005), and Harvey, Liu, and Zhu (2016) address inference under searches across alternatives; Bailey and López de Prado (2014) address selection and distributional distortions in Sharpe-ratio evaluation. [L06] Their inclusion here establishes methodological context, not a claim that those procedures were run. The reported capstone inference uses its documented permutation, bootstrap, and Bonferroni procedures. A family correction cannot repair

unavailable inputs, incomplete trial accounting, or an unsuitable resampling assumption; a Sharpe-specific adjustment is not automatically an adjustment for rank IC.

Data and scope of the historical record

The original study uses company-year observations for 81 BIST firms over 2020-2025. Its reported cohort contains 40 firms. The original feature set contains 40 columns covering statements, valuation, price history, momentum, and the benchmark; the target is the vendor's dated calendar-year next-year return in nominal TRY. [N06] These descriptions identify a frozen historical design. They do not establish that all firms were eligible throughout the period or that the panel represents a historical index population.

The PIT-v2 report distinguishes its evaluable panel from its held-out test cross-sections. For fiscal years 2020-2024 it records 323 published rows, of which 308 are evaluated under the primary specification; 11 lack a quote at prediction and 4 are excluded for a one-day move beyond ±25 % within the target window. [N07] These are report-level panel counts, not a claim that all evaluated rows are out-of-sample tests. The test cross-sections are FY2022, FY2023, and FY2024, containing 78, 78, and 79 firms, respectively. [N08] No aggregate test count or new eligibility calculation is added here.

Market-value availability is uneven. The frozen report records 164 rows valid on a consistent adjustment basis; 27 excluded for disagreement between share records and split history; 5 excluded for an unexplained close jump; and 123 unavailable because no share count exists for training-only firms. [N09] These categories describe market-value treatment and must not be conflated with whole-row evaluation exclusions. Missing stored values remain null; this manuscript neither reconstructs missing facts nor makes a new missing-data choice. The correction operates on the admitted features and rows specified in the historical protocol. [P01]

Original design and the audit

The original evaluation used expanding-window walk-forward prediction in calendar test years 2023-2025, three rank baselines, and six machine-learning models: OLS, ridge, lasso, elastic net, random forest, and gradient boosting. [N10] Predictions were evaluated using within-year Spearman IC, pooled as an equal-weighted mean across test years. The inference record uses within-year permutation tests and bootstrap intervals, with Bonferroni adjustment over the machine-learning family. The original walk-forward was not preregistered. [N11]

The preserved equal-weight baseline IC is +0.150 with permutation $p = 0.017$. The original machine-learning results range from −0.153 for random forest to +0.093 for ridge; the corresponding Bonferroni p-values are 0.11 and

0.94. [N12] These figures are historical outputs. Their preservation does not retain the original positive-baseline interpretation. Nor does the historical machine-learning null excuse contaminated evaluation: a defect may invalidate the information set even when it does not produce a statistically positive model.

The publication audit found that 31 of the 40 features used full fiscal-year statements before they were public. The results record describes normal reporting lags of 60-70 days, with the FY2023 deadline extending to 20 May 2024 and SASA's FY2022 deadline to 10 May 2023. [N13] A return window beginning in January therefore overlapped information arrival. The problem was not that the statements were assigned to the wrong accounting year; it was that the evaluation treated the end of that year as their availability date.

The audit also identified mixed adjustment bases: historical share counts were multiplied by back-adjusted prices when constructing market value and related valuation ratios. In addition, 14 labelled rows preceded the security's first quote, and five shared a vendor placeholder return of 56.947991 %. [N14] These historical audit counts refer to the original labelled panel. They need not equal the differently scoped exclusion counts in PIT-v2. A duplicate placeholder is a measurement warning, not evidence of a legitimate realised return. None of these defects is repaired in this drafting pass; we report the existing correction and its boundaries.

Availability-aware protocol and guard

PIT-v2 defines availability as the time information became public, rather than the end of its observation period. Each admitted model feature has a registry entry recording its source, period, availability rule, transformation, and adjustment basis. The documented guard rejects values dated after prediction, unregistered features, and training labels realised too late for the split. The protocol and existing guard tests are reproducibility artifacts; this manuscript does not claim to have rerun those tests. [P01]

The primary evaluation uses STANDARDIZED_CONSERVATIVE_CUTOFF: statements are treated as public from the applicable statutory deadline. The uniform cutoff is 31 May of the following year, and prediction and entry occur on the first trading day afterward. Exact filing dates exist for only 6 company-years, so actual-filing-date mode was not run on the full universe. [N15] The standardized rule is a labelled approximation. It does not establish an observed publication timestamp for every statement, and an unobserved late filer could still be assigned availability too early.

Price features use information available by the cutoff. The target uses Yahoo adjusted closes from the first trading-day close after the cutoff to the last trading-day close on or before the next cutoff. The protocol requires earlier training labels to be realised before the next prediction. Market value is used on a consistent price-share basis where supported and otherwise excluded;

pre-listing and corporate-action-defect rows are excluded under the registered rules. These are the historical protocol's measurement choices, not claims that a generic vendor adjustment fully resolves corporate actions. [P01]

The test predictions begin on 1 June 2023, 3 June 2024, and 2 June 2025 for FY2022, FY2023, and FY2024, respectively. [N16] Moving the entry date changes the economic question as well as the availability condition. The corrected design evaluates a later return window and uses a different return source and row set. It therefore supports an availability-aware reassessment of the historical claim, while leaving the isolated contribution of each correction unidentified.

An executable check is useful because it converts a documented temporal rule into a condition that can reject an inadmissible run. Its assurance is conditional on the input timestamps, source completeness, and rule semantics. A guard that correctly compares two inaccurate dates remains scientifically insufficient. Software correctness and historical-information validity are therefore separate obligations.

Corrected findings and sensitivity interpretation

The primary equal-weight baseline reports IC +0.031, permutation p = 0.63, and a 95 % bootstrap interval of −0.100 … +0.162. Its within-year ICs are +0.145 for FY2022, −0.051 for FY2023, and −0.000 for FY2024. [N17] We reproduce the reported precision, including rounded negative zero; we do not reinterpret it as a measured negative effect. The interval includes both signs and is not an equivalence bound. A non-rejection does not establish that the true association is zero.

The secondary results below reproduce the frozen report. All Bonferroni-adjusted p-values in this family are 1.00. [N18]

Model	Pooled IC	Permutation p	Bonferroni p
OLS	+0.073	0.27	1.00
Ridge	+0.001	0.99	1.00
Lasso	+0.056	0.39	1.00
Elastic net	−0.021	0.75	1.00
Random forest	−0.021	0.75	1.00
Gradient boosting	+0.028	0.68	1.00

Table source: [N18]. These are secondary associations in a post-hoc design, not a ranking of deployable strategies or a basis for a new model-selection exercise.

The statement-only comparison is informative but also bundled. The diagnostic using statements before publication reports +0.193 (p = 0.002), while the PIT-v2 statement-only sensitivity reports +0.041 (p = 0.54). [N19] The use of the same fiscal-year statements weakens an explanation based solely on using older accounting periods. Nevertheless, the return window, source, and row set also change. The observed pattern is consistent with publication-time look-ahead; it does not estimate an isolated timing effect, an attribution share, or the contribution of a particular code fix.

The registered sensitivities also describe how alternative admitted specifications behave. The earlier deadline-month cutoff gives baseline IC +0.102 (p = 0.12); removing market-value features gives +0.044 (p = 0.50); price features alone give −0.004 (p = 0.95). [N20] They are descriptive checks, not candidates from which to select a replacement headline. Their variation leaves both publication proximity and sample limitations relevant to interpretation.

In the statement-only sensitivity, random forest reports IC −0.176, with unadjusted p = 0.007 and within-specification Bonferroni p = 0.041. Across the four sensitivities, however, the report identifies 24 machine-learning tests and states that this result does not survive correction over that family. [N21] We retain that qualification and make no inverse-strategy claim. The isolated adjusted value cannot be promoted by ignoring the wider search. This manuscript adds no statistical test or recomputed family adjustment.

Cohort selection: established mechanism, measurement pending

The capstone universe-81 firms, 40 reported-was selected in 2026 from then-current listings under an undocumented rule. [N01] Firms already delisted by the selection date cannot enter that frame, so historical coverage is structurally conditioned on survival. An index exit alone is not a delisting, and this mechanism does not prove that every historical index leaver is missing. Whether the original selection also filtered or ranked firms using realised returns is **UNKNOWN**. The source label “winner cohort” is insufficient to establish that procedure. [C01]

Selection-replication placeholder - NOT RUN. A comparison of the frozen design on the rule-based, survivor-inclusive point-in-time universe U1 is planned under FIQ-17. Its source contrast would compare the capstone cohort under alternative source data; its selection contrast would compare U1 with the capstone cohort under a common source. Neither contrast is an available result. Selection-effect direction, magnitude, uncertainty, and performance bounds remain **UNKNOWN**. There is no numerical entry to impute, and no source-effect or selection-effect estimate is inserted here. [C02]

The available coverage comparison is narrowly cross-sectional. At 2020-10-01, an existing security-code projection contains 100 codes; 60 overlap the

capstone training list and 40 do not. [N22] These are literal code identities at one date. The non-overlapping codes are not established delistings; the comparison is neither a historical annual membership series nor an issuer-equivalence analysis. It does not admit a research formation or measure a performance effect. Annual membership, overlap, and exit-classification fields remain null in the supporting disclosure. Future work will use predeclared point-in-time eligibility and follow firms through their holding or event windows after index exit or delisting. [C01]

Power, controls, and the meaning of the null

The reported analytic minimum detectable |IC| is 0.185 at 80 % power for a design described as three years × 78 firms. [N23] This is a sensitivity characteristic of the reported design, not a confidence limit on the actual effect, and not a claim that the realised cross-sections are identical in size. It cannot establish the absence of smaller signals or be turned into a universal threshold for financial predictability.

Historical positive controls injected synthetic signal into the published harness, with 200 repetitions per level. The recorded detection rates are 0 % at IC 0.1, 17 % at 0.2, 62 % at 0.3, and 93 % at 0.4. Negative controls using permuted targets and noise, with 1,000 repetitions each, produced false-positive rates of 2.8 % and 2.6 % at a 5 % level. [N24] These are already-published control summaries; no simulation is performed for this manuscript. The synthetic injections test a harness under specified conditions and are not fabricated source observations or evidence of an investable signal.

The controls also illustrate a separation of obligations. A permutation control can behave conservatively while the unpermuted dataset violates publication timing. The historical defect-injection record explicitly reports that the original guards did not detect timing leakage; the availability-aware guard addresses timestamps. [N25] Thus a passing software or false-positive check cannot establish that every feature was public. Nor do these reported controls validate every dependence assumption in the corrected financial panel. Their proper role is to bound what the evaluation can detect and what specific failures its checks can expose.

Registration, prior exposure, and limitations

The original walk-forward was not preregistered. The correction was designed after the contaminated result and audit diagnostics were known, then registered before its own outcome computation in the amendment committed at 079bb832b8f9466ab70ae6ef4d60c02cd3080875. [P02] This chronology is more informative than a binary registered/unregistered label. Prospective commitment relative to the corrected run constrains its interpretation; it does not erase earlier outcome exposure or convert an audit-driven specification into independent confirmation.

Development ends at 2022-12-31; 2023-01-01 through 2025-12-31 remains the locked period under the adopted charter. Capstone outcomes in that period have already been observed, so the historical holdout is semi-clean. [G01] Reading the expressly authorized frozen aggregate reports for drafting does not open other protected data. Clean forward evidence requires documented prior-exposure and cross-study checks in addition to post-registration publication dates. The previously exposed forward registration remains a dated record; it is not silently repaired or represented as clean. [P02]

Several limitations remain material. The standardized availability rule substitutes deadlines for a complete filing-time census. It can delay usable information in ordinary years and can assign a late filing too early. The small set of test cross-sections and the high-inflation nominal-TRY setting limit precision and external validity. Mixed-source and adjustment uncertainties remain relevant even after explicit exclusions. A market-value coverage restriction can change which observations inform valuation inputs; it does not by itself establish missingness independent of the target. [N26]

The jump exclusion uses a price movement inside the target window. The corrected result is therefore conditional on retrospective row admission under the historical rule; it is not an evaluation of a wholly ex ante implementable trading screen. This additional selection issue must remain visible alongside the attempt to quarantine adjustment defects. We preserve the registered rule and its reported results rather than silently redefining the evaluated population. [N07]

Inference is inherited from the frozen study. Within-year permutations and ticker resampling do not automatically resolve sectoral dependence, common market shocks, or dependence across years. This manuscript does not add a dependence-robust test or a comprehensive reconstruction of the historical trial family. Those gaps limit inferential strength and prevent a new confirmatory label. A corrected null on a survivor-conditioned cohort is also not evidence that survivor-inclusive results would be null.

The audit supports withdrawing the old predictive interpretation and reporting the corrected estimates with their power limits. It does not establish that fundamentals are uninformative in every market, that all learning methods fail, or that an economically usable strategy exists here. Rank association alone would not establish net trading performance, capacity, transaction-cost robustness, or an investment edge. Selection measurement, exact-filing sensitivity, and independent scientific review remain outstanding.

Reproducibility and availability

The reproducibility record preserves original and corrected findings rather than overwriting one with the other. The results summary and exact PIT-v2 report are the sole outcome-number sources used for this draft. The protocol, amendment, and accompanying crosswalk distinguish methods, chronology, numeric evidence, and unknowns. Crosswalk entries pin a full commit and source lines so that subsequent editing cannot silently change the evidence cited.

The frozen report points to machine-readable manifests containing input hashes and environment details. Those manifests and other outcome files were not opened in this drafting pass. Reproduction must use the pinned historical inputs: rebuilding adjusted-price panels from a later fetch can change the series. The manuscript's static checks establish traceable transcription and document consistency; they do not certify a fresh successful reproduction, source-rights clearance, public availability of every input, or scientific validity. [P03]

Stored missing facts remain null, historical records remain intact, and no study is executed to complete the narrative. The selection placeholder is part of the evidence boundary, not a result awaiting a preferred sign. Its eventual replacement requires separately authorized, registered work and complete reporting whatever the findings show. A reader can assess the completed audit without assuming that missing membership or performance evidence exists.

Conclusion

This case shows how a financial prediction exercise can preserve chronological model fitting while violating the information set available at prediction. An availability-aware correction removes the original positive-baseline interpretation in the reported design, but its bundled changes prevent causal attribution to publication timing alone. Survivor-conditioned sample construction remains a separate look-ahead problem whose performance effect has not been measured. The reproducible contribution is the connection between an inspectable historical record, explicit temporal contracts, executable rejection of inadmissible inputs, and claims proportionate to power, exposure, and selection evidence. The appropriate conclusion is a bounded negative finding and an unfinished selection audit, without a claim of established investment advantage.

References used in this draft

Bibliographic identities come from the verified subset of references.bib at the preparation commit; exact keys, source locations, and content-access limits are in the literature crosswalk. No unresolved PBO pagination is used. The bibliography is not a systematic-review claim.

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